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High-Pressure Synthesis and Thermal Conductivity of Semimetallic θ-Tantalum Nitride

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The lattice thermal conductivity (κ_{ph}) of metals and semimetals is limited by phonon-phonon scattering at high temperatures and by electron-phonon scattering at low temperatures or in some systems with weak phonon-phonon scattering. Following the demonstration of a phonon band engineering approach to achieve an unusually high κ_{ph} in semiconducting cubic-boron arsenide (c-BAs), recent theories have predicted ultrahigh κ_{ph} of the semimetal tantalum nitride in the θ-phase (θ-TaN) with hexagonal tungsten carbide (WC) structure due to the combination of a small electron density of states near the Fermi level and a large phonon band gap, which suppress electron-phonon and three-phonon scattering, respectively. Here, measurements on the thermal and electrical transport properties of polycrystalline θ-TaN converted from the ε phase via high-pressure synthesis are reported. The measured thermal conductivity of the θ-TaN samples shows weak temperature dependence above 200 K and reaches up to 90 $Wm^{-1}K^{-1}$, one order of magnitude higher than values reported for polycrystalline ε-TaN and δ-TaN thin films. These results agree with theoretical calculations that account for phonon scattering by 100 nm-level grains and suggest κ_{ph} increase above the 249 $Wm^{-1}K^{-1}$ value predicted for single-crystal WC when the grain size of θ-TaN is increased above 400 nm.

1. Introduction

Materials with high thermal conductivity (κ) are useful for improving heat dissipation and lowering the hot-spot temperatures in microelectronic devices and many other technologies. In insulators and semiconductors, phonon-phonon scattering limits the lattice thermal conductivity (κ_{ph}) that dominates the total κ. In metals, the large electron concentration often leads to a larger electronic thermal conductivity (κ_e) than κ_{ph}. In semimetals, the electron concentration lies in an intermediate range that is insufficient to produce a large κ_e compared to κ_{ph}. In both metals and semimetals, electron-phonon scattering can provide another limiting mechanism for κ_{ph} at low temperatures or in structures with weak phonon-phonon scattering, such as tungsten[1] and some transition metal carbides.[2] One exception is graphite, a semimetal with a low electron concentration that does not prevent its basal-plane κ_{ph} from reaching 2000 $W\ m^{-1}\ K^{-1}$ at room temperature.[3] This value is close to the record-high κ_{ph} reported for diamond[4] and slightly higher than the κ_{ph} measured recently in isotopically purified cubic boron nitride (c-BN).[5] The ultrahigh κ_{ph} in these three materials systems is caused by the strong covalent bonding of light elements in the simple crystal structures.[6]

By comparison, semiconducting cubic phase boron arsenide (c-BAs) achieves a similarly ultrahigh κ_{ph} via a different route.[7–9] The presence of both light boron (B) and heavy arsenic (As) elements results in a large energy gap between the acoustic (A) phonon branches and the optical (O) phonon polarizations. This large A-O gap and the close bunching of the acoustic branches suppress three-phonon scattering[10] to yield a high κ_{ph} where four-phonon processes play an unusually important role.[11] These same phonon band features also yield high intrinsic carrier mobilities by restricting electron-phonon scattering and make BAs an emerging III-V semiconductor for high-performance electronics.[12–14]

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A number of binary compounds are made of a light element and a heavy one similar to BAs. However, κ_{ph} has been found or predicted to be low or modest in a number of them that are either metals or semimetals with frequent electron-phonon scattering, despite a large A-O gap and weak phonon-phonon scattering. One example is niobium carbide, where κ_{ph} is lowered by electron-phonon scattering over one order of magnitude at temperatures as high as 1000 K.[2] Meanwhile, hexagonal-phase semimetallic tungsten carbide (WC) exhibits a relatively high room-temperature κ_{ph} value, which is measured to exceed 100 W m^{-1} K^{-1} in polycrystals[15] and predicted to reach 249 W m^{-1} K^{-1} in isotopically purified single crystals due to a large A-O gap and relatively low electron density of states (EDOS) near the Fermi energy.[16]

In comparison, recent theoretical analysis[17] and first-principles calculation[18] have suggested the feasibility of achieving κ_{ph} approaching 1000 W m^{-1} K^{-1} at room temperature in hexagonal θ-phase tantalum nitride (θ-TaN) in the WC structure due to a unique combination of a large A-O gap and a very low EDOS in this semimetal made of light nitrogen (N) and heavy tantalum (Ta). Besides the θ phase that is stable upon high-pressure and intermediate-temperature synthesis, the phase diagram of TaN includes additional low-pressure hexagonal ε phase and a metastable high-pressure and high-temperature cubic δ phase. Past measurements have obtained room-temperature κ values of 9.5 W m^{-1} K^{-1} and 3.2 ± 0.2 W m^{-1} K^{-1} for polycrystalline ε-TaN and TaN films, respectively.[19,20] Used as a diffusion barrier in metal interconnects in microelectronics and in magnetoresistive random access memories,[21–23] the TaN thin films consists of a mixture of both crystalline δ phase and amorphous phases.

There have been prior reports of high-pressure synthesis of θ-TaN via solid-phase reactions from hexagonal ε-TaN powder.[24–26] It has been suggested that displacement of the Ta atom occupying a site in the trigonal plane likely causes the formation of θ-TaN under high pressure and high temperature.[26] Single-phase bulk samples of θ-TaN grown by static compression at 0.8-3 GPa were found to be stable at high temperatures between 1773 and 2073 K.[25] A belt-type high-pressure apparatus was also used to prepare θ-TaN at 2273 K under pressure in the range between 2.0 and 5.5 GPa.[26]

Here we report on the high-pressure synthesis and transport property measurements of θ-TaN polycrystals with sub-micron grain sizes. Three different measurement methods give κ values that vary among the different samples in the range between 15 W m^{-1} K^{-1} and 90 W m^{-1} K^{-1} at room temperature and is insensitive to temperatures above 200 K. At room temperature, the samples appear to be a semimetal with a *p*-type carrier concentration. Although the measured κ of different samples show some weak correlation with the sub-100 nm grain size and carrier concentration, spatial variation in the same sample is observed and obscures such correlations. Nevertheless, both the observed κ magnitude and temperature dependence agree with the first principles-based calculations that account for phonon scattering by 100 nm level grain sizes and obtain 940 W m^{-1} K^{-1} and 755 W m^{-1} K^{-1} at 300 K along *a* and *c* axes, respectively, for single-crystal θ-TaN.

2. Results

2.1. Microstructure of As-Synthesized θ-TaN

Three TaN pellets with a WC-type hexagonal crystal structure (space group $P\bar{6}m2$; no. 187) were grown under a high pressure from ε-TaN powder as the starting material.[25–27] Each synthesized pellet was cut into bar shapes for the transport properties measurements (see Methods). In the following, each bar-shaped sample is labelled as S*nl*, where the number *n* indicates the pellet number, and the letter *l* is used to differentiate the different bar-shaped samples cut from one pellet.

Figure 1(a) compares the X-ray diffraction (XRD) patterns of the starting ε-TaN powder and five samples obtained from the

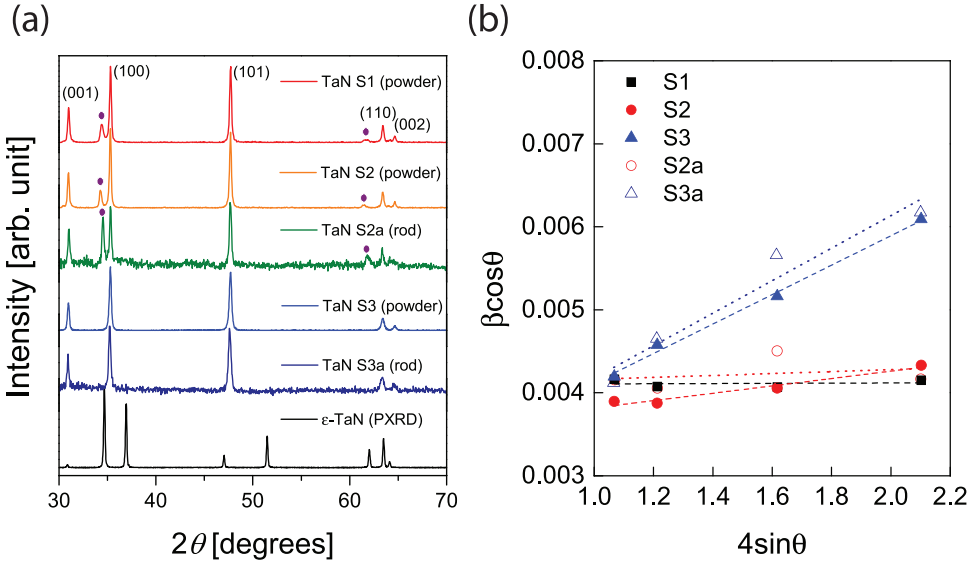


Figure 1. a) XRD patterns of the three as-grown crystals (TaN) and source powder (ε-TaN) in either pulverized powder form (S1,S2,S3) or in the original rod shape without the pulverization step (S2a,S3a). The indexed peaks are a signature of the θ-TaN phase, while the peaks from impurity phases are marked with a purple dot. b) The W–H analysis of pulverized (filled symbol) and rod (open symbol) TaN samples. The strain is extracted from the slope and the crystalline size is extracted from the *y*-intercept of the fit. Dashed and dotted lines are fitting lines of pulverized and rod TaN samples, respectively.

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Table 1. P-T conditions for sample synthesis, lattice parameters, averaged grain size (D) extracted from the W–H analysis, and the unitless strain factor (ϵ).

Sample	P [GPa]	T [K]	Time [min]	*a* [nm]	*c* [nm]	*D* [nm]	ϵ [10^{-4}]
S1	4.2	1800	30	0.294	0.290	33.8 (powder)	0.1
S2	4.2	2000	30	0.295	0.290	41.1 (powder) 34.3 (bar, S2a)	4.4 (powder) 1.2 (bar, S2a)
S3	4.2	1500	15	0.294	0.290	59.3 (powder) 62.5 (bar, S3a)	17.7 (powder) 19.6 (bar, S3a)

high-pressure synthesis. Among the five samples, S1,S2,and S3 were prepared by pulverizing part of a pellet into powder, while S2a and S3a are the bar-shape samples prepared for transport property measurements. The XRD patterns for all the five samples contain clear peaks that match those of θ-phase TaN. **Table 1** shows that the synthesis conditions and the calculated lattice parameters extracted from the XRD patters are within the range of other reported experimental values.[25,26,28] In addition, ε-TaN peaks marked with purple dots in Figure 1(a) are observed in the XRD profiles of S1 and S2 but not in S3. To better understand the origin of the ε-TaN peaks observed in the XRD of S1 and S2 powdered samples, we further extracted tens of micron-size particles from the interior of synthesizes products from the S1 and S2 batch for measurement with rotation-sample XRD (RSXRD) (Figure S1, Supporting Information), which does not reveal the ε-TaN peaks. These XRD patterns from powdered samples, rod samples, and interior particle samples suggests that S1 and S2 have inhomogeneous ε-TaN distribution in the high-pressure products, while S3 shows no evidence of remaining ε-TaN phase.

XRD can be utilized to evaluate the average crystallite size and lattice strain due to dislocation from broadening of the diffraction peaks. XRD patterns were refined to provide the lattice parameters and averaged crystallite size using Williamson–Hall (W–H) analysis[29,30] with the uniform deformation model according to the following equation,

$$\beta \cos\theta = \frac{k\lambda}{D} + 4\epsilon \sin\theta \quad (1)$$

where β is the corrected peak width at half-maximum intensity (FWHM, radian), θ is the peak position (radian), k is a shape factor equal to 0.9, λ is the wavelength of the radiation (1.541 Å for CuKα radiation), D is the average crystallite size in nanometers, and ϵ is the strain induced due to crystal imperfection and distortion. The obtained D from the XRD data varies between 33 and 63 nm based on the XRD results (Figure 1(b) and Table 1). Each sample batch showed similar D values regardless of pulverization.

Electron backscatter diffraction (EBSD) was performed on the polished surface. Most of the examined areas have low pattern quality except for some micron-size grains (Figure S2, Supporting Information). The result suggests that the sizes of most grains are comparable to the EBSD resolution and support the average crystallite size obtained from W–H analysis of XRD results. The polished sample surfaces were further treated with diluted hydrofluoric acid (HF) for 20 min prior to scanning electron microscopy observation of the grain boundaries. The SEM images (**Figure 2**) reveal a wide range of grain sizes from tens of nanometer to micrometer scale, in agreement with the EBSD observation of a mixture of grain sizes.

Figure 3 shows the Annular Dark Field (ADF) Scanning Transmission Electron Microscopy (STEM) images of S2a and S3a together with a low-magnification STEM image of S3a from cross-sectional lift-out lamella prepared using a gallium focused ion beam system. The lattice parameters from the ADF-STEM images are consistent with the XRD results. The Fast Fourier Transform (FFT) patterns of each region can be indexed based on the lattice structures of θ-TaN. The low-resolution TEM images reveal a range of grain sizes similar to the SEM results.

2.2. Impurities Analysis of θ-TaN

The relative impurity concentration variation as a function of depth among two representative samples (S2a,S3a) is measured by time-of-flight secondary ion mass spectrometry (TOF-SIMS) depth profiling. **Figure 4** shows the depth profiles of several impurity species of interest that have been normalized to the corresponding TaN profile of each sample to account for variations in the sputtering beam current and sample surface morphology.[31,32] The spike at the very surface (d = 0) results from the matrix effect.[33] The main impurities are related to carbon and oxygen in both

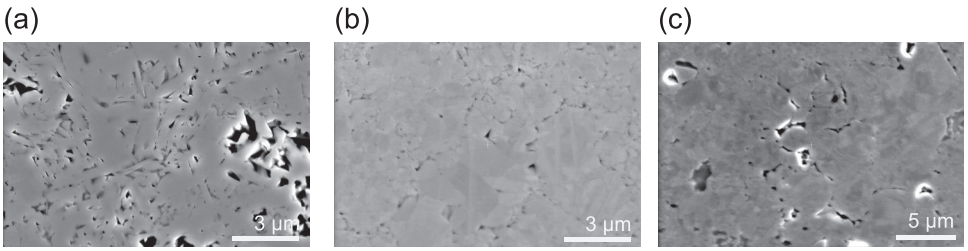


Figure 2. SEM images of the polished surfaces followed by acid treatment (a,b: S2a,c: S3a).

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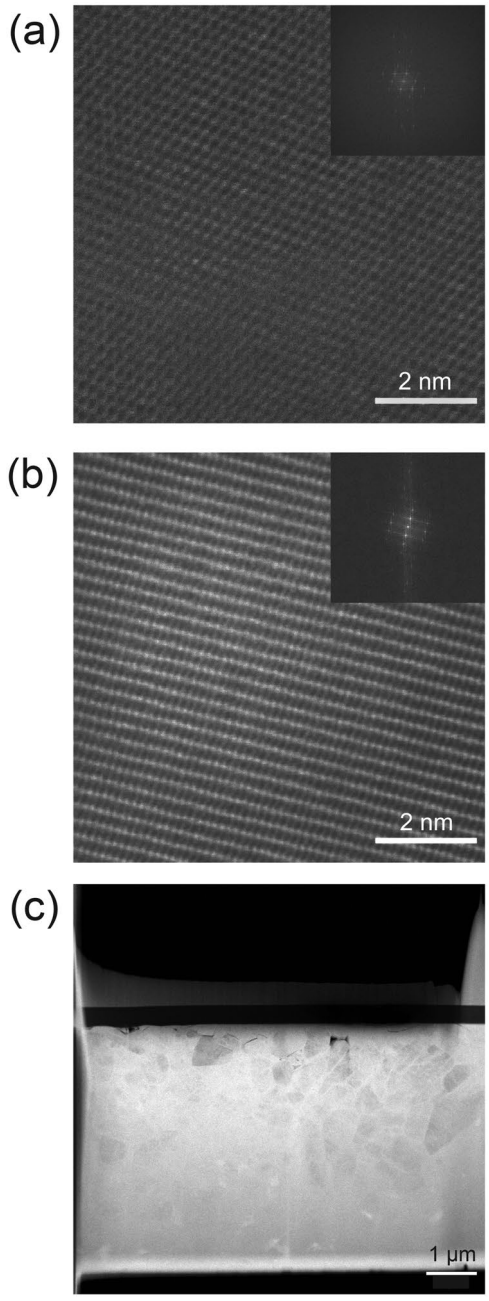


Figure 3. Annular dark-field STEM images taken within one grain of the a) TaN sample S2a along the [001] zone axis and b) TaN sample S3a along the [100] zone axis. The insets in (a) and (b) show the FFT patterns of the STEM image. c) Low-magnification STEM image of sample S3a.

samples. As a common impurity in the nitrides,[34] the oxygen impurities (O^-, O_2^-, TaO^-, TaO_2^-, BO_2^-) concentration in S2a is larger than those in S3a at a depth beyond a few hundred nanometer where the secondary ion (SI) yields reach a steady state. On the other hand, sample S3a shows more carbon impurities (C^-, CN^-) than sample S2a. Due to the lack of calibration standards for TaN, however, the absolute impurity concentration cannot be determined from the TOF-SIMS signals.

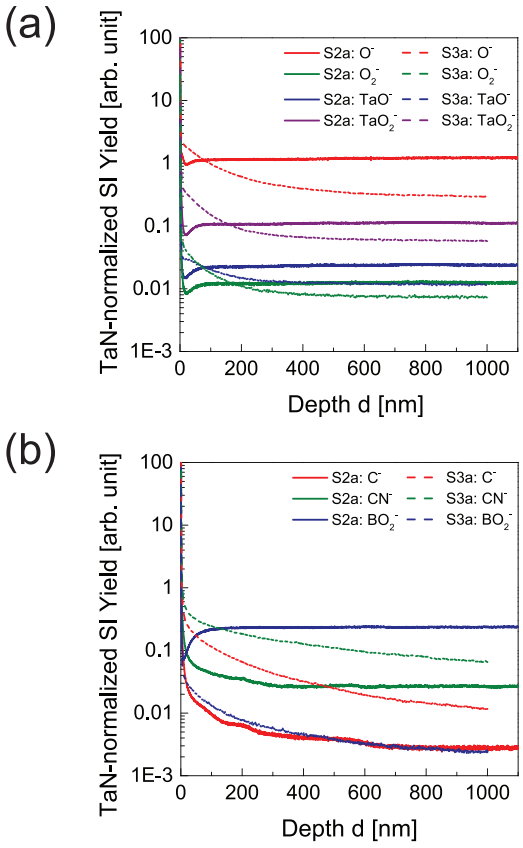


Figure 4. Depth dependence of a) O^- related and b) C^- related or BO_2^- impurity SI yields normalized by the TaN^- yield for the two samples measured by TOF-SIMS. Solid lines and dashed lines indicate SI yields from S2a and S3a, respectively. The depth profile of S2a measured up to 1100 nm while that of S3a measured up to 1000 nm, due to the difference of the sputtering rate of Cs^+ sputtering beam.

2.3. Thermal Conductivity of θ-TaN

Figure 5(a) presents the temperature-dependent κ values of θ-TaN samples from the three different batches (S1,S2,and S3). The κ values of these samples were measured using three different methods based on differential thermocouples (TC), differential thin film resistance thermometers (RTh), and time-domain thermoreflectance (TDTR). The measured κ values at room temperature vary from 15 W m^{-1} K^{-1} to 90 W m^{-1} K^{-1}. After S1b and S2a were measured by the TC method, these two bar samples were polished for further RTh measurements, which obtained similar results for the same sample as those from the TC measurements.

In addition, both the TC and RTh methods show that the measured κ values are insensitive to T above 200 K in the six samples. This feature indicates that phonon-phonon scattering is not the dominant scattering mechanism compared to extrinsic scattering by grain boundaries and impurities in these polycrystals at room temperature and below. Among the six samples, the highest and similar κ values are obtained from TC measurements of sample S2b and RTh measurements of sample S3a, the latter of which shows lower oxygen impurity but higher C impurity concentration than sample S2a with a lower κ.

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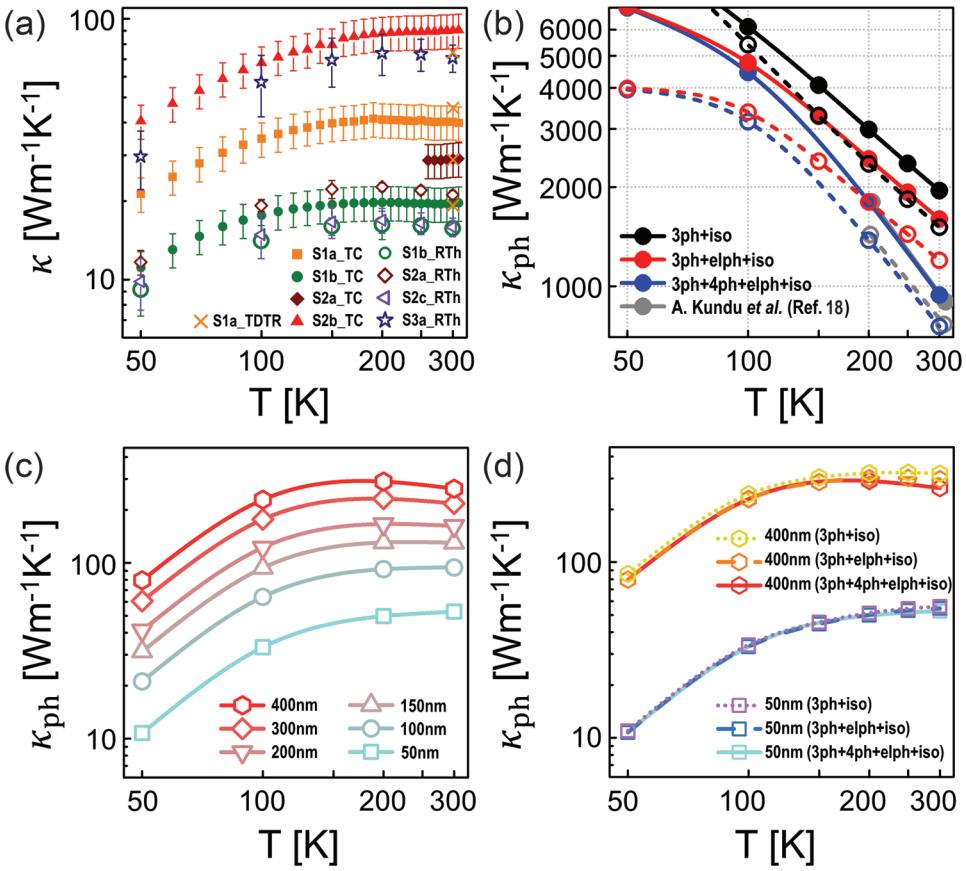


Figure 5. a) Temperature dependence of the measured κ of θ-TaN samples by the TC method (filled symbols), RTh method (open symbols), and TDTR method (crosses). The error-bars indicate the propagated uncertainty from the individual measurements and additional systematic errors caused by the geometric errors. b) Temperature dependence of calculated intrinsic κ_{ph} in bulk single-crystal θ-TaN along the *a* (solid lines, filled symbols) and *c* (dashed lines, open symbols) axes, respectively. Different line colors are used to show the results of calculations that account for different combinations of the three-phonon (3ph), isotope-phonon (iso), four-phonon (4ph), electron-phonon (elph) scattering processes, and the reported calculation (Ref. [18]). c) Calculated temperature dependence of κ_{ph} with different grain size when all phonon scattering mechanisms are considered. Symbols are calculation results, which are fitted with the lines. d) Calculated temperature dependence of κ_{ph} at two different grain sizes (50 and 400 nm) when different combinations of phonon scattering mechanisms are considered.

However, the same TC method obtained quite different results for two different samples cut from the same pellet, including S1a and S1b from batch S1 and S2a and S2b from batch S2. TDTR was further used to investigate the spatial variation of the local κ in sample S1a.[35] At room temperature, the local κ values vary from 16 to 75 W m^{-1} K^{-1}. These results reveal large variation of the crystalline quality not only between different samples from the same batch but also between different locations of the same sample. This variation makes it difficult to observe a clear dependence of the κ on either the grain size or impurity concentrations.

To understand the temperature dependence of the measured κ, we have calculated the κ_{ph} of θ-TaN by accounting for different scattering mechanisms. By considering third nearest neighbor interatomic force constants (IFCs) in the calculation of four-phonon scattering, the calculation obtains 940 Wm^{-1} K^{-1} and 755 Wm^{-1} K^{-1} at 300 K along *a* and *c* axes, respectively, for single-crystal θ-TaN (Figure 5(b)). These values are close to the previous report that accounts for up to the second nearest neighbor.[18] In addition, we have calculated the κ_{ph} of θ-TaN with different grain sizes to understand the effect of phonon-grain boundary scattering (Figure 5(c)). When the grain size is reduced below 300 nm, the calculated κ_{ph} shows a similar temperature dependence as the measured data. When the grain size is reduced to the range between 100 nm and 50 nm that is slightly larger than the average values extracted from the PXRD data, the calculated room-temperature κ_{ph} is reduced to the range between 52.7 Wm^{-1} K^{-1} and 94.1 Wm^{-1} K^{-1} (square and circle in Figure 5(c)) comparable to the upper and intermediate ranges of the measurement results.

We investigate the effect of defect scattering due to mass disorder and strain that is revealed by the XRD measurement results. For a 400 nm grain size, increasing the strength of point defect scattering to be three orders of magnitude higher than natural isotope impurity scattering (g_{iso}) reduces the calculated κ_{ph} at 300 K by 7% (Figure S3, Supporting Information). In comparison, such increase shows negligible effect on κ_{ph} when the grain size is reduced to 50 nm.

We further investigate the effects of electron-phonon scattering and four-phonon scattering in single-crystal and polycrystalline θ-TaN. For the single-crystal calculation results in Figure 5(b), inclusion of electron-phonon scattering reduces the

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calculated κ_{ph} at 300 K by 18% and 27% along the *a* axis and *c* axis, respectively, much smaller than the 230% reported for WC,[16] whereas inclusion of four-phonon scattering reduces the calculated κ_{ph} at 300 K by 41% and 37% along the *a* axis and *c* axis, respectively. In comparison, inclusion of electron-phonon and four-phonon scattering yields very small and negligible effects on the calculated κ_{ph} when the grain size is 400 and 50 nm, respectively, as shown in Figure 5(d). This comparison reveals a dominant effect of grain boundary scattering compared to electron-phonon and four-phonon processes for these polycrystalline samples with grain sizes >400 nm, consistent with the calculated scattering rates for different mechanisms (Figure S4, Supporting Information). The grain size needs to be increased into the micrometer scale for electron-phonon and four-phonon processes to be apparent in the κ_{ph}.

2.4. Raman Spectra of θ-TaN

Prior Raman measurements[36] of δ-TaN have observed defect-mediated first-order acoustic (A) phonon band centered near 150 cm^{-1}, a broad first-order optical (O) phonon band near 550 cm^{-1}, and a small second-order O-A band near 448 cm^{-1} corresponding to the difference between an optical mode and an acoustic mode. In comparison, the Raman spectra measured on our θ-TaN samples S2a and S3a (**Figure 6**) contain an apparent band centered near 250 cm^{-1}, a relatively sharp peak at 480 cm^{-1}, a small and broad bump near 545 cm^{-1}. Based on our calculation of the phonon dispersion (Figure S5(a), Supporting Information) and the symmetry of the zone-center optical phonon modes, the 574.6 cm^{-1} mode is IR active, whereas the 545.9 cm^{-1} mode is both Raman and IR active and is close to the observed feature near 545 cm^{-1}. Assigned here as defect-mediated first-order acoustic band and second-order O-A band, respectively, the other two bands are observed at higher frequencies, 250 cm^{-1} and 480 cm^{-1}, than their counter parts in δ-TaN. Moreover, the observed O-A band in θ-TaN is much sharper than that in δ-TaN.

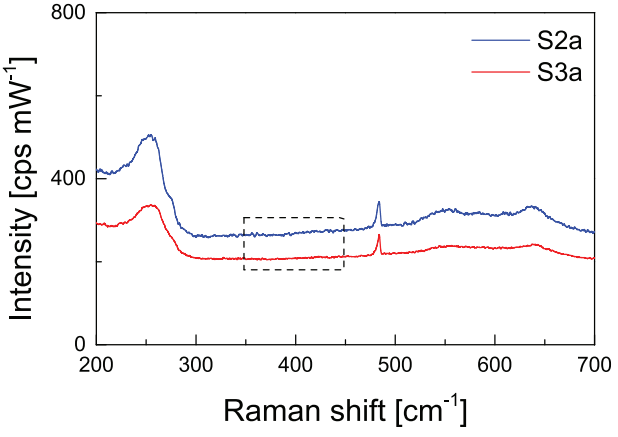


Figure 6. Measured Raman spectra of S2a and S3a at room temperature. The black dashed rectangle indicates the 350–450 cm^{-1} phonon frequency range, which is inside the 295.5–504.3 cm^{-1} A-O phonon band gap and used to calculate the electronic Raman scattering background intensity.

However, the observed Raman linewidth does not show a simple correlation with the phonon lifetime in metals and semimetals where electronic Raman scattering (ERS) occurs due to the interaction between localized electronic and vibrational defect states. Point defects, including N site vacancies, play an essential role in not only the observation of the first-order acoustic Raman band away from the zone center, but also ERS background over a large frequency spectrum inside and above the A-O gap. The ERS background can obscure the Raman-active optical band near 545.9 cm^{-1} and broaden the O-A band in both δ-TaN and θ-TaN. In prior works, the normalized ESR background intensity has been correlated with the defect concentration and κ_{ph} of BAs[9,31,32] and MoS_2.[37] Here, we integrate the background intensity in the 350–450 cm^{-1} frequency range inside the A-O gap and normalize it with the O-A peak height. The normalized values, 7.816 cm^{-1} and 7.818 cm^{-1} for sample S2a and S3a, respectively, are similar. Hence, the higher κ measured in S3a is likely mainly due to the larger grain size (Table 1) than S2a, although spatial variation of the sample quality makes it difficult to rule out the effect of different defect concentrations.

2.5. Electrical Properties of θ-TaN

Existing band structure calculations have suggested that θ-TaN is a semimetal.[17] Semimetals are a class of materials with characteristics between those of semiconductors and metals. Although they lack an energy gap, their intrinsic carrier density can vary widely, between 10^{15} cm^{-3} and 10^{20} cm^{-3}, based on the band overlap and the size of the carrier pockets.[38] These values are much lower than those of metals, which are usually on the order of 10^{23} cm^{-3}. They are similar to, and in some cases lower than, the 10^{18}–10^{20} cm^{-3} values of heavily doped semiconductors. However, semimetals often show far greater Hall mobility (μ_H) than metals and strongly doped semiconductors. Consequently, the electrical conductivity (σ) of semimetals is comparable to that of heavily doped semiconductors. Note that the μ_H is significantly reduced in heavily doped semiconductors as a result of ionized impurity doping, and in metals as a result of electron-electron and electron-phonon interactions.

Hall measurements of the polycrystalline θ-TaN samples reveal hole-like majority carriers with a concentration (p_H) that is relatively insensitive to temperature (Figure S6, Supporting Information). The p_H values are in the range of the aforementioned intrinsic carrier density of semimetal, as shown in **Figure 7**(a). Among them, sample S1b exhibits the highest hole concentration. This finding suggests that a large doping concentration is likely another factor of the observed lower measured κ of this sample than the other. Phonon scattering by both defects and charge carriers can decrease the κ_{ph}. The free carriers in θ-TaN are caused by impurities, which can decrease κ.

The measured electrical properties of θ-TaN are summarized in Figure 7(b). The measured σ values are of the order of 10^5 S m^{-1}, which are more than one order of magnitude smaller than the theoretical values for θ-TaN single crystals.[18] Above 20 K, the σ of θ-TaN decreases slightly with increasing temperature, showing a typical metallic behavior, where

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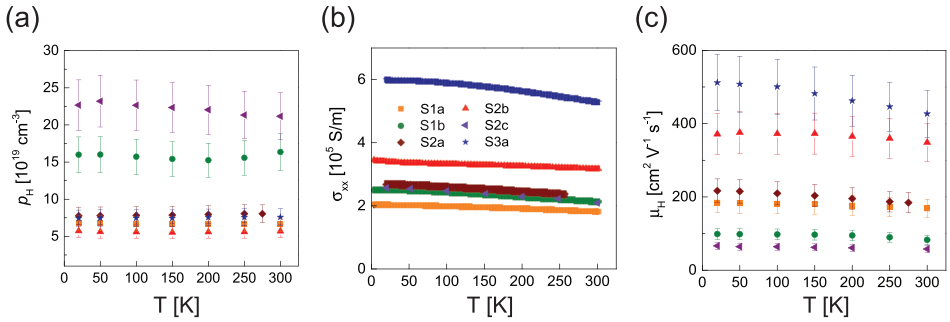


Figure 7. Temperature dependence of measured a) hole concentration (p_H), b) longitudinal electrical conductivity (σ_{xx}), and c) Hall mobility (μ_H) of θ-TaN samples. The error-bars indicate the uncertainty from the individual measurements and additional systematic errors caused by the geometric errors. The legend in panel (b) is for all panels.

electron-phonon scattering increases the resistivity with temperature.

Similarly, the obtained Hall mobility (μ_H) (Figure 7(c)) slightly decreases with increasing temperature, revealing a combined effect of scattering processes by phonons, impurities, and grain boundaries. The measured μ_H are not as large as reported electron mobility (μ_e) of typical semimetals, such as 3.5×10^4 cm^2 V^{-1} s^{-1} in HgTe[39] and 1.2×10^4 cm^2 V^{-1} s^{-1} in Bi.[40] Nevertheless, they are larger than that of the *p*-type doped Si at the same p_H.[41] The sample with the lowest p_H exhibits the highest μ_H, as its defect concentration is the lowest.

The room-temperature thermoelectric power *S* values of the θ-TaN samples and other room-temperature physical properties are listed in **Table 2**. Due to the higher p_H of actual samples, the measured *S* values are a factor of 3 to 4 smaller than the reported maximum of calculated *S* value of 32 μV K^{-1} at 300 K.[42] All samples show positive *S* at all measured temperature range (Figure S7, Supporting Information), consistent with the Hall measurements. *S* decreases with increasing p_H, as expected in other semimetals.[38]

The electronic contribution to κ according to the Wiedemann-Franz law varies from 2.6% to 11.8% of the total κ, with the lower and higher values found for sample S2b and sample S2c, respectively. These ratios are comparable to the previous theoretical calculation.[18] The S1b and S2c showed the highest percentage contributions from the electronic contribution due to their high p_H, which helps to increase the electronic contribution to κ and reduces the lattice contribution via enhanced phonon-defect and phonon-electron scattering.

3. Conclusion

In summary, high pressure has been used to synthesize polycrystalline semimetallic θ-TaN samples with the grain size that vary in the range between tens of nanometers up to 1 μm in the same sample with an average value of 33–63 nm based on PXRD measurements. Multiple thermal transport measurement methods have obtained similar κ values, in the range between 15 W m^{-1} K^{-1} and 90 W m^{-1} K^{-1} at room temperature, for these polycrystalline θ-TaN samples. The measured magnitude of κ and its weak temperature dependence agree with theoretical calculation results, which account for phonon-grain boundary scattering based on the observed average grain size as well as the large phonon band gap and small EDOS. The measured κ shows some correlation with the grain sizes for two samples. In addition, the lowest κ value appears in a sample with the highest hole concentration p_H, indicating possible additional effects of defect scattering. However, spatial variation in the same sample is observed and makes it difficult to establish a clear correlation between the measured κ with either the grain size or the p_H. While these values are still one order of magnitude smaller than the theoretical value for θ-TaN single crystals due to phonon-grain boundary scattering and phonon-defect scattering, they are an order of magnitude larger than those reported for polycrystalline hexagonal ε-TaN and TaN thin film with mixed crystalline δ-TaN and amorphous phases. The theoretical calculations suggest that the κ of polycrstalline θ-TaN can increase above 300 W m^{-1} K^{-1}, which exceeds the predicted single-crystal value of semimetallic WC, when the grain

Table 2. Room-Temperature Physical Properties of θ-TaN samples. The temperature is 275 K for the electrical conductivity and Hall measurement data of sample S2a and 300 K for other data and other samples.

Sample	κ [W m^{-1} K^{-1}]	κ_e/κ	σ [10^5 S m^{-1}]	*S* [μV K^{-1}]	p_H [10^{19} cm^{-3}]	μ_H [cm^{-2} V^{-1} s^{-1}]
S1a	40.3 (TC)	0.034	1.85	10.4	6.6	168.5
S1b	19.6 (TC) 15.7 (RTh)	0.081 0.101	2.17	6.9	16.4	82.8
S2a	29.0 (TC) 21.1 (RTh)	0.052 0.071	2.41 (275 K)	9.3	8.0 (275 K)	184.4 (275 K)
S2b	90.3 (TC)	0.026	3.19	13.8	5.7	348.6
S2c	15.8 (RTh)	0.118	2.57	7.3	21.2	58.0
S3a	70.8 (RTh)	0.048	5.27		7.6	426.9

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size is increased above 400 nm. Further increase of the grain size and reduction of the defect concentration may lead to high κ reaching up to 900 W m^{-1} K^{-1} and provide another example of the band engineering approach to high κ.

4. Experimental Section

High-Pressure Synthesis: High-pressure syntheses of the TaN crystals were performed at 4.2 GPa and between 1500 K to 2000 K with a Cubic-type multianvil module (Rockland Research Co.). The ε-TaN powder (Alfa Aesar, 99.5%) was pressed into a pellet and sealed in a hBN crucible. Each crucible was contained in a graphite heater surrounded by a $LaCrO_3$ sleeve for thermal insulation. Cr-doped MgO cubic cell were used as the pressure medium. The schematic illustration of high-pressure cell assembly was shown in Figure S8(a) (Supporting Information). The cell was compressed to the target pressure and then heated at a rate of 15 K min^{-1}. After heating for 30 min, the sample was quenched to room temperature with the power of the heater turned off. As-synthesized cylindrical crystals (Figure S8(b), Supporting Information) were obtained upon the disassembly of the cubic cell. The diameter of the sample varies from 2.7 mm to 3 mm while the sample height ranges from 3.6 to 4.4 mm.

X-Ray Diffraction: Bruker AXS Smart APEX2 diffractometer (Rotation-Sample XRD, RSXRD, Mo Kα radiation) and Rigaku Miniflex 600 Diffractometer (Powder XRD, PXRD, Cu Kα radiation) were used to determine the phase of the obtained samples at room temperature. The XRD profiles of the products were acquired in two distinct ways for comparison. First, the RSXRD was used to examine the homogeneity of micrometer-sized samples that were collected from a small particle of the pellet. The monochromatic X-ray was focused on the samples, and the imaging plate detected the diffracted X-rays. Using the standard analysis program, the Debye rings acquired by the detector were integrated into intensity -2θ data. After preparing the bar-shaped samples for the electrical and thermal measurements, the bar-shaped samples and pulverized powders of remaining products were used for PXRD analysis. The corrected FWHM, β, is calculated as

$$\beta = \left(\mathrm{FWHM}_{\mathrm{meas}}^2 - \mathrm{FWHM}_{\mathrm{inst}}^2\right)^{\frac{1}{2}} \quad (2)$$

where $\mathrm{FWHM}_{\mathrm{meas}}$ was measured FWHM of peaks, $\mathrm{FWHM}_{\mathrm{inst}}$ was the instrumental broadening of PXRD. The $\mathrm{FWHM}_{\mathrm{inst}}$ was calculated by Caglioti function[43]

$$\mathrm{FWHM}_{\mathrm{inst}}(2\theta)^2 = \mathrm{U}\tan^2\theta + \mathrm{V}\tan\theta + \mathrm{W} \quad (3)$$

where U,V,W were 3.1×10^{-3}, 0, 1.1×10^{-2}, respectively.[44]

Microstructure Analysis: The morphology and microstructures of the samples were examined with a field-emission scanning electron microscope (FE-SEM; ZEISS Neon 40). The polished sample surfaces for EBSD analysis were prepared with finer diamond lapping films (9,3,0.5,0.1-micron grade), and electron backscatter diffraction (EBSD; Scios 2HiVac Dual beam FIB/SEM System) measurements were carried out. The single spot probe measurement was performed to evaluate the pattern quality with conditions of 1×1 binning (i.e., no binning), 400–420 ms exposure, and a gain of 0, followed by the EBSD mapping with conditions of 2×2 binning, 20 ms exposure, and a gain of 10. In addition, 10% diluted hydrofluoric acid (HF) was used to etch the sample surface to observe grain boundaries. The cross-sectional TaN lamella for the STEM study were prepared via a standard lift-out procedures with a final polishing step at 5 kV and 77 pA to reduce surface damage using the same FIB/SEM system. Annular dark-field (ADF) STEM images in Figure 3 were acquired with a JEOL NEOARM equipped with a probe corrector for STEM operated at an accelerating voltage of 200 kV and a beam current of 10 µA. A dwell time of 10 µs was used for imaging.

Thermal Conductivity Measurements: With the use of a steady-state differential thermocouple based comparative approach (TC)[7] and a differential thin film resistance thermometry method (RTh),[31,45] the κ of the samples was measured in the temperature range from 20 to 300 K. For the thermocouple-based approach, bar-shaped θ-TaN samples were cut from as-grown samples and then polished with diamond lapping films. The reference was a cylindrical Constantan of 0.25 mm in diameter and 2 mm in length. Plasma-enhanced chemical vapor deposition (PECVD) was used to deposit 300 nm of SiN_x dielectric on the surface of bar-shaped θ-TaN to electrically insulate the thermocouple from the semimetallic sample. The SiN_x-coated θ-TaN sample was linked in series with the Constantan reference using silver epoxy. As shown in Figure S8(c) (Supporting Information), they were suspended and positioned between a resistive heater and a Cu heat sink. The differential thermocouple attached to the θ-TaN was made of a Constantan wire connected to two Cu wires at both ends. Two additional Cu wires with a diameter of 12.7 µm were attached to the Constantan reference. The thermocouple junctions were attached to the surface of the sample with silver epoxy, while the lead wires were suspended in vacuum. The sample was encased in a Cu radiation shield and placed in a cryostat that had been evacuated to 1×10^{-6} torr using a turbomolecular pump. The two differential thermocouples monitor the temperature drops along the Constantan reference and the θ-TaN sample. At temperatures between 150 and 300 K and for a Si calibration sample with comparable dimension and room-temperature κ, the κ value obtained by the TC method was well within 10% from the literature value.[7]

In addition, the RTh method was used to validate the thermocouple measurement results of samples with a sufficiently large size for patterning thin film resistance thermometers.[31,45] For this method, bar-shaped θ-TaN samples measuring $0.4 \times 0.1 \times 2$ mm^3 with a 300 nm thick SiN_x layer were prepared. An Au/Cr serpentine thin film heater was patterned on a separate silicon bar. The silicon bar with the heater was joined to one end of the sample using silver epoxy, while the other end of the sample was attached to a copper heat sink by silver epoxy (Figure S8(d), Supporting Information). As the sample length was insufficient for patterning both the heater and thermometers on the θ-TaN sample surface, this separate thin film heater was required to maintain a sufficient distance between the two thermometers patterned on the small sample in order to achieve a measurable temperature drop and 1D temperature distribution. Due to the use of microfabrication to form the heater and thermometers directly and precisely on the sample bar, this method was particularly appropriate for mesoscopic samples with lateral dimensions ranging from several hundred micrometers to a few millimeters. Two measurements with different numbers of Al:Si bonding wires to the heater were used to directly determine the heat loss from the heater into the bonding wires. In addition, the bonding wires of thermometers were positioned outside the sample region of interest. Measurements were conducted in a cryostat with two radiation shields to decrease the surface heat loss. A differential Wheatstone bridge circuit was used to directly measure the slight difference in temperature between two thermometers. With the heating current stepped to 16 discrete steady values that follow a sinusoidal pattern over 20 cycles, FFT was used to calculate the second harmonic modulation amplitude of the temperature response to the Joule heating to enhance the signal-to-noise ratio. Monte Carlo error propagation was used to accounts for the measurement errors in the temperature drop, temperature coefficients of resistance (TCR), and sample dimensions.[31,45] For a similarly small Si calibration sample at temperatures above 20 K, the κ values from a pure heat conductivity model and the radiative fin model analysis of the RTh data was within 6.2%. The κ values obtained by the radiative fin model of the RTh measurement of the Si calibration sample were within 10% and 5% difference from the literature values at temperature 400 and 100 K, respectively.[45]

For samples that were too small for thin film differential thermometry measurements, they were polished for TDTR measurements (Figure S8(e), Supporting Information). TDTR was an optical pump-probe technique which can be used to measure the κ of both bulk and nano-structured

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materials.[35,46] The pump beam periodically heats the sample surface while the probe beam detects the temperature of the sample surface via thermoreflectance. A 10 X objective with a spot size of 5.5 um (radius) and a modulation frequency of 9.3 MHz in the TDTR measurements was used. Using diamond lapping films, samples were polished to have smooth surfaces with a root-mean-square surface roughness (S_q) of less than 20 nm at a 10×10 μm^2 area, and the S_q of samples at various sites was measured using an optical profilometer (Keyence VK-X1100 Optical Profilometer).

Raman Measurements: The 532 nm wavelength laser with the power of 10 mW were used as the excitation. The Raman laser beam was focused by 50× achromatic objective lens onto the multiple location of two samples (S2a and S3a) and the spot size was less than 1 μm. Each acquisition takes 5 min with the grating size of 2400 mm^{-1}, and all Raman spectrum acquired in room temperature by iHR320 Horiba Spectrometer with Synapse CCD.

Electrical Properties Measurements: Electrical measurements have been carried out on the same θ-TaN samples used for κ measurement. The resistivity (ρ) and the Hall coefficient data were collected using the Physical Property Measurement System (PPMS) from Quantum Design in the temperature range between 20 K and 300 K and the magnetic field sweeping between ± 3 T to obtain carrier concentration (p_H) and mobility values (μ_H). The thermoelectric power (S) was measured in the temperature range 200–300 K with a home-made setup. The Seebeck coeffcient of the copper block used as the metal lead for S measurements is 6.5 μV K^{-1}.

Time-Of-Flight Secondary Ion Mass Spectrometry (TOF-SIMS): In order to probe the impurities in the TaN crystals as function of depth, an ultrahigh-resolution TOF.SIMS 5 instrument configured with 500 eV Cs^+ sputtering and 30 keV Bi/Mn analysis ion beams from IONTOF GmbH was used. The analysis chamber was kept at ultrahigh vacuum (UHV) at a pressure below 2×10^{-9} mbar. All detected secondary ions of interest were negatively charged. The 500 eV Cs^+ sputtering beam was used for layer removal with a typical sputtered area of 300×300 μm, while the sample composition was mapped by the 30 keV Bi/Mn ion beam raster scanning an area of 100×100 μm. The sputtering rate of Cs^+ sputtering beam was estimated at 0.06 and 0.056 nm s^{-1} for sample S2a and S3a, respectively.

Computational Details: First-principles calculations based on density functional theory (DFT) were executed by using the Quantum ESPRESSO materials simulation suite.[47] The local density approximation to DFT, and norm-conserving pseudopotentials from the Pseudo Dojo library was employed.[48] In all calculations a planewaves kinetic energy cutoff of 70 Ry was employed, and total energies down to 10^{-10} Ry per unit cell and forces down to 10^{-8} Ry/bohr was converted. The optimized lattice parameters of TaN in the space group $P\bar{6}m2$ (no. 187) structure were $a = b = 2.898$ Å and $c = 2.843$ Å.

The 2nd-order, 3rd-order, and 4th-order IFCs were calculated via finite differences in supercells using the codes PHONOPY,[49] thirdorder,[50] and fourthorder,[51] respectively. Interactions up to the sixth nearest-neighbor and up to the fourth nearest-neighbor for calculating third-order and fourth-order IFCs, respectively were considered. All IFCs were calculated using a $5 \times 5 \times 5$ supercell sampled via a $2 \times 2 \times 2$ k-point grid. Convergence tests of the κ with respect to the number of nearest neighbors in both the third- and fourth-order IFCs was performed (Figure S9, Supporting Information).

Phonon lifetimes resulting from electron-phonon scattering were calculated using the EPW code.[52] To obtain maximally localized Wannier function, initial projections corresponding to d orbitals on Ta and s and p orbitals on N was chosen. electrons, phonons, and electron-phonon matrix elements on coarse $8 \times 8 \times 8$ k- and q-point grids and perform Wannier interpolation to dense k- and q-point grids up to $84 \times 84 \times 84$ and $16 \times 16 \times 16$, respectively was evaluated. In the current EPW version, the phonon lifetimes were obtained using the standard Fermi golden rule formula, which contains the square moduli of the electron-phonon matrix elements was noted. This was the same approach followed in previous report.[19] A more refined approach would require replacing this term by the product of a screened electron-phonon matrix elements and bare matrix elements, see Sec. V.A.3 of Ref. [53] for a detailed discussion.[53] As a result of this choice, we expect our electron-phonon scattering rates to be somewhat underestimated. A more detailed analysis of this effect will be presented elsewhere.

The total phonon scattering rates were obtained by combining three-phonon, four-phonon, electron-phonon scattering, isotope scattering, and phonon-boundary scattering rates via Matthiessen's rule, and the κ_{ph} was obtained by solving the self-consistent Boltzmann transport equation (BTE) for phonons as implemented in the ShengBTE code.[50,54,55] The FourPhonon[51] module was used as an extension module of ShengBTE, which was modified appropriately to include electron-phonon scattering rates from EPW. All κ_{ph} were calculated on a $16 \times 16 \times 16$ q-point grid. The justification for this grid can be found in supplement material of Ref. [18].

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

electron-phonon coupling, high pressure syntheses, phonons, semimetal, thermal conductivities

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